



New understanding of the voltammetry of platinum in non-adsorbing electrolytes

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ABSTRACT

The present study uses cyclic voltammetry to investigate the "butterfly-shaped" response assigned to the adsorption of hydroxyl species (OHads) on Pt(111) in acidic media. We present new voltammetric profiles recorded in methanesulfonic acid (CH₃SO₃H, MSA) solutions and mixtures of MSA and perchloric acid with striking features. Previous studies have suggested that the response of OH adsorption on Pt(111) depends on the degree of water structuring through the formation of hydrogen-bonded networks and the orientation around the electrolyte ions. Under this hypothesis, structured water would produce a broad peak, whereas unstructured water would be responsible for the sharp peak. In this study, increasing the concentration of the methanesulfonate anion in a 0.10 M perchloric acid solution resulted in a negative shift of the broad peak alongside a positive shift of the sharp peak, suggesting that the methanesulfonate anion has a notable capacity to structure water. The direct interaction of anions with OHads through hydrogen bonding could also participate in the splitting of the OH adsorption region. Additionally, a new OH adsorption response, evidenced by the appearance of new voltammetric peaks at 0.55 and 0.85 V was observed with an increase in methanesulfonate anion concentration (from 0.10 to 0.30 M). It was also found that perchlorate anions in the presence of methanesulfonate anions facilitated the formation of those peaks with a synergic effect greater than that of the individual anions. Analysis of the voltammetric charges and their dependence on pH and anion concentrations suggests that the potential dependence of OH adsorption is redistributed depending on the effect of those external factors on the structure of water, but its total amount remains essentially constant ($q_{0.5-0.9 \text{ V RHE}} \sim 96.4 \mu\text{C cm}^{-2}$). The effect of very small amounts of sulphate and chloride anions on the shape of the voltammogram has also been investigated. The former has a structure-making effect, shifting the sharp voltammetric peak to higher potentials, whereas the latter has a structure-breaking effect, shifting this peak to lower potentials. These shifts agree with previous studies that used perchloric acid as the electrolyte. Following the previous suggestions, it is proposed here that different voltammetric peaks in the OH adsorption region correspond to OH adsorption from different water sources, differing in the degree of hydrogen bonding, and that the methanesulfonate anion has a remarkable ability as a hydrogen bond maker. Other explanations, such as MSA-specific adsorption or the direct interaction of the anion with OHads are also considered, but regarded as less likely to account for all the experimental observations reported in this work.

1. Introduction

Cyclic voltammetry has become one of the most basic tools for the fundamental study of electrocatalytic processes [1–3]. With the variation of scan limits, scan rate, and concentration of electroactive species, it is possible to gain a rather complete understanding of the phenomena

that characterise the electrochemical behaviour of a given electrocatalytic material. Such an understanding must always start from the comprehension of the phenomena that occur in the supporting electrolyte in the absence of any other electroactive compounds (i.e. the voltammetric response of the blank electrolyte). Under these "blank" conditions, the cyclic voltammogram is typically characterised by the

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presence of capacitive currents and electrosorption (pseudocapacitive)-associated peaks.

Within the framework described above, the voltammetry of Pt(111) in acidic media can be considered a milestone in the progress of interfacial electrochemistry [4]. This voltammetric profile was first described in 1980 after the introduction of the flame annealing methodology for the decontamination and ordering of a well-defined single-crystal surface [5]. Initial interpretations of the Pt(111) voltammetry were controversial [6], but it later became widely accepted that such a profile can be described as composed of several regions [4,7]. At low potentials, close to the onset of hydrogen evolution, the voltammogram is characterised by a broad peak, almost a plateau of current, owing to the hydrogen adsorption/desorption process. At higher potentials, anion adsorption occurs. In sulfuric acid solutions, sulphate adsorption takes place in the positive scan just at the end of hydrogen desorption [8,9]. In contrast, in perchloric acid, another archetypical electrolyte, anion adsorption takes place just below the onset of the oxidation of the surface, and it is separated from the hydrogen adsorption by what is typically considered a purely capacitive region, usually designated as the double-layer region. The voltammetric profile above 0.5 V is composed of two signals, a broad feature followed by a rather sharp peak. These processes are very reversible, and the voltammetric profile is perfectly symmetrical against the x-axis, conforming to what has been named in the literature as “the butterfly” [4,10]. As mentioned above, the interpretation of “the butterfly” was controversial at the beginning [4,6,11], but it was later shown that it corresponds to anion adsorption [12,13]; usually, it has been accepted that it is due to hydroxyl (OH_{ads}) adsorption [14,15].

One of the strongest arguments for assigning the butterfly to hydroxyl adsorption and not to perchlorate adsorption is the fact that essentially identical voltammograms are obtained for perchloric acid and trifluoromethanesulfonic acid [15]. The hypothesis is that both anions are non-specifically adsorbed on the surface; hence, the butterfly corresponds to the electrosorption of the only common anionic species in both electrolytes, namely, the hydroxide species (formed directly from water). Hence, if the adsorption peaks come from water splitting, their characteristics should depend on the water structure which, in turn, is influenced by electrolyte anions. Other non-adsorbing electrolytes include hydrofluoric and methanesulfonic acids. However, hydrofluoric acid is dangerous and difficult to handle because it attacks glass, and voltammograms in fluoride are usually taken at a higher pH [16,17]. While the methanesulfonate anion has also been considered as a non-adsorbing anion, subtle changes in the voltammetric profile take place in the presence of this species [18].

In this study, we present non-previously described voltammetric profiles recorded in methanesulfonic acid (CH₃SO₃H, MSA) solutions, and perchloric acid (HClO₄) and MSA mixtures. These voltammograms challenge some of the established explanations previously described and call for a reinterpretation of the cyclic voltammogram or, at the very least, reconsideration of the role of the anion in the electrosorption processes that characterise the voltammetric profile.

2. Experimental

Single-crystal platinum electrodes were prepared according to previously described procedures [7,19]. Before each experiment, the electrode was annealed in propane gas + air flame and cooled in a hydrogen + argon (1:3 ratio) atmosphere. Afterwards, the electrode surface was protected with a droplet of ultrapure water saturated with these gases and transferred to the electrochemical cell. The electrochemical cell and all glassware were cleaned by soaking the material overnight in a slightly acidic solution of KMnO₄. Then, the glassware was rinsed with an acidic solution containing diluted H₂O₂ to remove the traces of manganese species. The glassware was boiled using ultrapure water (18.2 MΩ cm, Elga Purelab Ultra) at least five times. The solutions were prepared from MSA (Merck for synthesis, ≥99 %), perchloric acid

(70–72 % Supelco, Merck), sulphuric acid (96 % Supelco, Suprapur, Merck), and NaCl (99.99 %, Suprapur, Merck) using ultrapure water (18.2 MΩ cm) and deaerated with argon (Ar ≥ 99.995 %) for at least ten minutes. The pH values were measured using a pH-Meter Crison Basic 20 (measuring error ≤0.01). Cyclic voltammetric profiles were recorded using an EG&G PARC 175 signal generator paired with an e-DAQ EA161 potentiostat and an e-DAQ ED410 recorder at a scan rate of 50 mV s⁻¹.

3. Results

The cyclic voltammograms obtained with a high-quality Pt(111) electrode in 0.10 M solutions of both MSA and HClO₄ are shown in Fig. 1A. These voltammograms are in good agreement with those reported in the literature [2,4,6,7,18,20]. In both cases, a hydrogen region was observed at low potentials, which was virtually identical regardless of the nature of the electrolyte anion. At ca. 0.4 V the current reaches a minimum that can be considered nearly capacitive, although small contributions from hydrogen desorption or anion adsorption in this potential range cannot be completely excluded [21]. At potentials above 0.6 V, the voltammetric current starts rising again. The CO displacement experiment unequivocally shows that this current is due to a net negative charge, as it would correspond to an anion oxidative electrosorption process [12,22]. The most common interpretation for this region of the voltammogram is that the current is due to OH_{ads} adsorption. While OH⁻ is present only at negligible concentrations at acidic pH, it can be formed on the electrode surface from the dissociation of water [4]. As shown in Fig. 1A, clear differences were observed between both acidic media in this region. Under the assumption that neither perchlorate nor methanesulfonate anions adsorb specifically, these differences have been previously attributed to the role of the anion affecting the structure of water that supplies the OH to be adsorbed on the electrode surface [18]. Under this hypothesis, the existence of two peaks, a broad one (OH_b) and a sharp one (OH_s) in the region between 0.55 and 0.85 V, is attributed to structured and unstructured water, respectively. Therefore, the shift in the OH_s peak in the presence of methanesulfonate can be attributed to its structuring-making effect.

Theoretical density functional theory (DFT) calculations have reported that MSA has intrinsic polarisability that acts as a nucleation centre for the surrounding water molecules, creating a solvation shell where the water molecule network is strongly hydrogen-bonded to MSA; this shell is the main stabilising factor of MSA [23]. Similar behaviour has been reported for the methanesulfonate anion [24]. Such hydrogen-bonded water molecules would be responsible for the OH_b peak, while water molecules with dangling O—H bonds would be responsible for the OH_s peak. According to this interpretation, the redistribution of charges between the OH_b and OH_s peaks, when the two different anions are compared (methanesulfonate vs. perchlorate), would be due to the different capacities of both anions to favour or break the structure of water. This interpretation is reinforced by the results of experiments in perchloric acid solutions spiked with very small amounts of other anions that are well known to have either structure-making or structure-breaking effects [14]. Namely, a small concentration of sulphate anions, in the range between 10⁻⁸ and 10⁻⁵ M, decreases and slightly shifts the OH_s peak to a higher potential, while small concentrations of chloride, in the same range, increase and shift the OH_s peak to a lower potential. Since chloride is well known to have a structure-breaking effect (ions that decrease the amount of hydrogen bonding), while sulphate has a structure-making effect (ions that induce more hydrogen bonding), the described shifts will be consistent with the assignment of the OH_s peak to the contribution of OH from unstructured water, while the OH_b peak would be due to adsorption of OH from structured water clusters, interacting through hydrogen bonds.

As shown in the black curve of Fig. 1B, increasing the HClO₄ concentration from 0.10 M (black curve of Fig. 1A) to 0.30 M did not cause significant changes in the voltammetric profile. The onset of OH adsorption slightly shifts to lower potentials and the charge between the

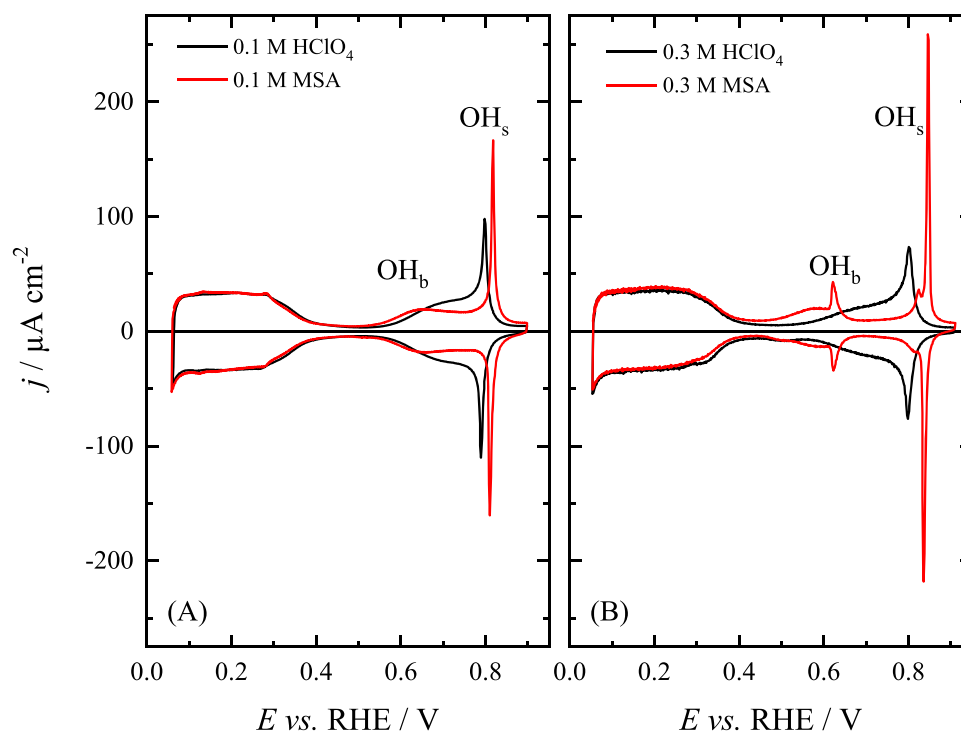


Fig. 1. Comparison of the voltammetric profiles of Pt(111) in (A) black line: 0.10 M HClO_4 , pH 1.10 red line: 0.10 M MSA pH 1.10, (B) black line: 0.30 M HClO_4 pH 0.64, red line: 0.30 M MSA pH 0.66. Scan rate= 50 mV s^{-1} .

OH_b and OH_s peaks is slightly redistributed (see Figure S1 in the Supporting Information for a direct comparison of both voltammetric profiles and the integrated charges), but overall, the voltammetric shape remains essentially the same and has the same tendency as reported previously [25], up to a concentration of 1 M HClO_4 .

Conversely, large and unexpected changes are observed when MSA concentration is increased to 0.30 M. New sharp peaks can be observed at 0.65 and 0.81 V, in addition to the one already present in 0.10 M MSA,

which shifts from 0.82 to 0.83 V. The net charge is redistributed and kept essentially constant (see Figure S2 in the Supporting Information). The chemical origin of this strikingly new voltammetric profile is difficult to elucidate at this point, since the increase in MSA concentration has changed both the pH and the total anion concentration. At higher potentials ($E > 1.0 \text{ V}$), the peak typically attributed to adsorbed O (that starts with water dissociation to adsorbed OH) [26] has great similarity to the peak in 0.10 M HClO_4 (as well as in 0.30 M HClO_4 , see Figure S3 in

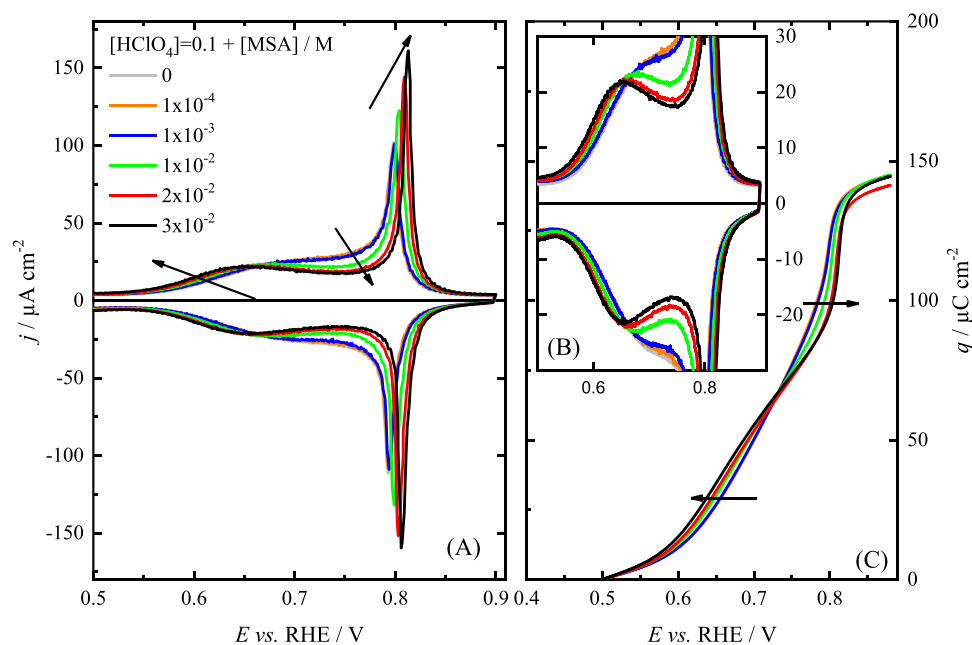


Fig. 2. Effect on the voltammetric profile of Pt(111) in 0.10 M HClO_4 from the addition of different concentrations of MSA, pH from 1.10 to 1.01. B) Enlargement of the OH adsorption region. C) The respective total charges as integrated from the voltammetric currents. Arrows indicate the direction of the shift with the increase in MSA concentration. Scan rate= 50 mV s^{-1} .

Supporting Information). This indicates that the effect of MSA is limited to potentials lower than 1.0 V.

To obtain further information on this new voltammetric profile, additional experiments were performed by selectively modifying the pH and concentration and mixing HClO₄ and MSA.

First, small amounts of MSA were added to the 0.10 M HClO₄ solution (Fig. 2A). For [MSA] < 10⁻² M, the pH (from 1.10 to 1.01) and ionic force are kept nearly constant. The addition of MSA progressively affects the OH adsorption region in the same way as previously described from the results of Fig. 1A (red curve); that is, the OH_b peak is displaced toward lower potentials, and the OH_s peak is shifted to higher potentials (in opposite directions). Fig. 2B shows an enlargement of the OH adsorption region. As in previous cases, the total charge is maintained constant, as clearly evidenced in the plot of Fig. 2C. While this result agrees with the interpretation given above that MSA has a water structure-making effect, in the range of MSA concentrations employed here, no new peaks are observed as those visible in 0.30 M MSA solution (Fig. 1B).

In a second experiment, the procedure was inversed: increasing amounts of HClO₄ were added to a 0.10 M MSA solution (Fig. 3). These results reveal a striking new feature: while at low HClO₄ concentrations, the voltammogram in MSA is virtually unaffected by the addition of perchlorate anions, when the concentration of HClO₄ is above 5 × 10⁻³ M, a pair of peaks at ca. 0.70 and 0.74 V are clearly discernible. These peaks can be considered equivalent to those observed at 0.61 and 0.85 V at high methanesulfonate concentrations (Fig. 1B). Such correspondence between both pairs of peaks will be clarified later by performing experiments in which the pH and MSA concentrations were varied.

In Fig. 3, MSA concentration is constant, and differences in the pH are negligible (from 1.10 to 1.01). Thus, the observation of these peaks in mixtures containing both anions points to a synergic effect at the heart of the specific ion effects phenomenon. To prove that the appearance of these peaks is not a mere effect of the slight change in pH, the voltammogram recorded in 0.11 M MSA (with no additional HClO₄) is shown in Figure S4, demonstrating that the increase of total acid concentration from 0.10 M to 0.11 M is not enough to develop these peaks, but the

addition of HClO₄ is required to observe them.

If the hypothesis described above is confirmed, that different voltammetric features are related to different degrees of water organisation, those peaks could be related to water strongly hydrogen-bonded to methanesulfonate and perchlorate anions. According to previous literature [27], in a mixed electrolyte, the effect of one anion on the solvation of a different anion is noticeable in the second solvation shell because solvation shells can bond to each other through hydrogen bonds, extending local interactions to larger distances. Thus, the first peak (Fig. 3B, labelled as OH_{b1}) could be associated with the formation of a first water solvation shell, whereas the second peak (OH_{b2}) could be associated with the second larger solvation shell. These results also indicate that the pair of peaks appears only after a certain hydrogen-bonded water saturation level is reached.

Although new peaks (OH_{b1} and OH_{b2}) emerged in the OH_b peak region (Fig. 3C), the OH_b and OH_s peaks in Fig. 3A remained essentially constant after perchlorate anion addition. It has been proposed that the presence of ions at a specific concentration does not alter the hydrogen-bond network outside the solvation shells [28]. This behaviour has been observed for perchlorate concentrations ranging from 10⁻⁶ to 10⁻² M. Therefore, this is consistent with the assignment of the OH_s to the contribution of OH from unstructured water. On the other hand, the structured water created by hydrogen bonds solvating methanesulfonate and perchlorate anions remained constant in the concentration range shown in Fig. 3. However, as shown later, small amounts of other anions such as chloride and sulphate have a visible effect on the position and magnitude of the OH_s peak, but not on the OH_b peak. Chloride and sulphate anions affect unstructured water, shifting the peak response to a lower potential with an increase in unstructured water for the former, and to a higher potential and a decrease in unstructured water for the latter. Also, the structured water created by hydrogen bonds solvating methanesulfonate and perchlorate anions remained constant in the concentration range shown in Fig. 3.

Fig. 2 shows that the addition of MSA to the HClO₄ electrolyte affects the positions of both the OH_b and OH_s peaks. Shifts in opposite directions could indicate that when the methanesulfonate anion

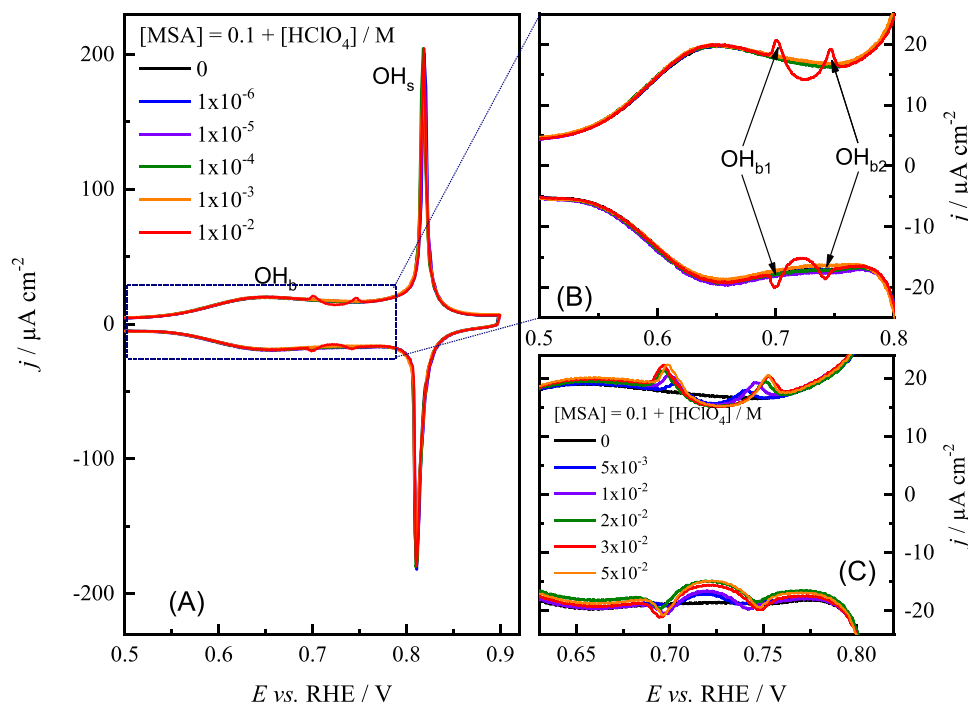


Fig. 3. A) Effect on the voltammetric profile of a Pt(111) in 0.10 M MSA of the addition of different concentrations of HClO₄, pH from 1.10 to 1.01. B) Enlargement of the OH adsorption region. C) Effect in the region between 0.60 and 0.85 V of further increments of HClO₄ concentration between 5 × 10⁻³ M and 5 × 10⁻² M. Scan rate = 50 mV s⁻¹.

concentration increases above 10^{-2} M (green curve), the number of hydrogen bonds yielding structured water increases and unstructured water decreases consequently. Therefore, it seems that the effect of ions on the structure of the water network may depend on the relative strength of the interactions forming the solvation shell. In contrast to Fig. 2A, Fig. 3A shows that the amount of structured/unstructured water remains constant for each small perchloric acid addition, until a specific concentration of 5×10^{-3} M (Fig. 3C) is reached. Above this concentration, two new peaks (OH_{b1} and OH_{b2}) appear in the region in which OH_b was previously dominant, which should be related to the rise of new adsorption states. Peak OH_{b1} at a lower potential could correspond to the lower (more negative) adsorption energy necessary to adsorb OH from the relatively strong hydrogen bond in the first solvation shell and closer to the surface. The strong hydrogen bond between the H_2O molecule and the oxygen of the anion may facilitate the rupture of the $\text{O}-\text{H}\cdots\text{OH}$ bond to form the OH moiety being adsorbed. On the other hand, peak OH_{b2} could correspond to weakly bonded hydrogen in the second shell subject to long-range interactions, requiring more energy to absorb OH from relatively weak hydrogen bonds and further from the surface. This assignment is done on the basis that adsorption of OH from structured water (OH_b) happens at lower potentials and as the structure is broken, adsorption of OH is shifted to higher potentials until the totally unstructured water happens under the peak OH_s . However, as shown in Fig. 4, a larger perchloric acid addition in the presence of methanesulfonate seems to have a long-range structural feature of hydrogen bond rearrangement dynamics that modifies the amount of structured/unstructured water because the peaks in the OH adsorption region shift in opposite directions.

Additional experiments were performed to study the displacement of the new peaks observed in the OH adsorption region with pH and methanesulfonate anion concentration. If the peaks were due to methanesulfonate adsorption, they should shift to more negative potentials

with an increase in its concentration and remain constant with pH changes [9]. However, if the peaks were due to OH adsorption, they should shift with pH variations and, in principle, should not be affected by methanesulfonate concentration, except through the indirect effect of this anion on the structure of water. It should be borne in mind that all potentials are measured vs. the RHE which has a Nernstian shift on pH. The results obtained in these experiments, however, display a non-Nernstian shift of the peak potential with pH (Fig. 4A and inset therein). First, Fig. 4A shows the results obtained in mixtures with a constant MSA concentration and increasing HClO_4 concentration. This experiment is a continuation of the one reported in Fig. 3A but with higher HClO_4 concentrations. As already indicated above, the presence of perchlorate induces the appearance of these new peaks, labelled as OH_{b1} and OH_{b2} , which shift in different directions: OH_{b1} shifts negatively, while OH_{b2} is displaced in the positive direction. Second, the peak labelled as OH_s also shifts in the positive direction, as shown in Fig. 4. The comparison of Figs. 1B and 4A (Figure S5 shows a direct comparison) shows that the peak labelled as OH_{b1} is similarly developed in both mixtures with a total acid concentration equal to 0.30 M, either 0.10 M MSA + 0.20 M HClO_4 or 0.30 M MSA. This observation reinforces the aforementioned idea of a synergic effect between the perchlorate and methanesulfonate anions. While in 0.30 M perchlorate, this peak is not observed, the presence of MSA and perchlorate anions triggers its appearance. This peak shifts to lower potentials as the pH decreases (more acidic). The shift is approximately 200 mV/decade (see inset in Fig. 4A). Although there is a large uncertainty in this number, owing to the narrow range of pH used for its calculation, the value is undoubtedly larger than the Nernstian shift of the RHE, 60 mV/decade. This contradicts the simple anion adsorption process as the origin of this peak. However, this would also be incompatible with a simple OH adsorption process, which should vary by exactly the same amount as the RHE and appear invariable in this reference scale. Hence, the shift is a clear

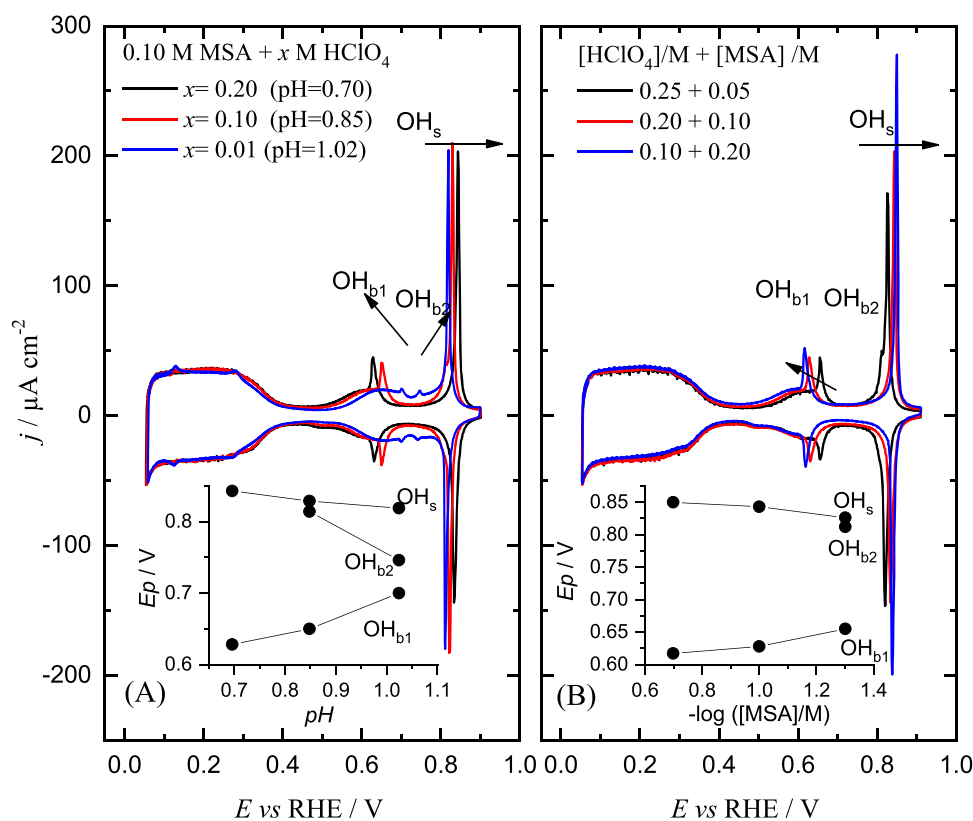


Fig. 4. Cyclic voltammetric profiles for Pt (111) recorded in A) 0.10 M MSA + x M HClO_4 (pH from 0.70 to 1.02) and B) different mixtures of MSA and HClO_4 with a total concentration at 0.30 M of acid (pH ca. 0.60). Scan rate = 50 mV s^{-1} . Insets in (A) and (B) show the shift of the position of the peaks (OH_{b1} , OH_{b2} , and OH_s) with pH and MSA concentration, respectively.

indication of the collaborative effect between the perchlorate and methanesulfonate anions. On the other hand, the peak labelled as OH_{b2} is visible for 10^{-2} M and 0.10 M perchloric acid concentrations, but it merges with the peak OH_s at a higher perchloric acid concentration (or lower pH). Finally, the OH_s peak does not shift with the addition of perchloric acid below 10^{-2} M, and when the concentration is above this limit, it shows a non-Nernstian shift with pH.

Fig. 4B presents data obtained in mixtures in which the total acid concentration is maintained at 0.30 M while varying the ratios between perchloric and MSA concentrations. Apart from the effect of different anions on the activity coefficient, the pH should remain constant in these experiments, which is ca. 0.60. In this case, both peaks, OH_{b1} and OH_s , shifted in different directions while increasing MSA concentration. Peak OH_{b1} shifted negatively as expected for an adsorption process linked to the methanesulfonate anion, while peaks OH_{b2} and OH_s shifted positively. The middle peak, OH_{b2} , merges with OH_s and it is only visible in one of the mixtures. Figures S6 and S7 in the Supporting Information show that although the net charge involved in these adsorption processes is redistributed across the different concentration mixtures, it remains essentially constant throughout the OH adsorption process.

To induce further modifications in the structure of the interfacial water, the effect of additional anions, such as chloride and sulphate, on the voltammetric profile was studied. As previously reported [14], in 0.10 M HClO_4 , chloride has a breaking effect on the structure of water and shifts the OH_s peak to lower potentials. Conversely, sulphate is reported to have a structure-making effect, and the OH_s peak slightly shifts to higher potentials. Here, 10^{-6} M H_2SO_4 was added to a mixture of 0.10 M MSA + 10^{-2} M HClO_4 (Fig. 5A). As can be seen, the peaks at ca. 0.70 and 0.74 V are suppressed, while the OH_s peak broadens and shifts to higher potentials. This last effect is similar to that observed in HClO_4 [14]; the shift of the OH_s peak towards higher potentials could be explained by the sulphate forming hydrogen bonds with the surrounding water, creating its own solvation shell and decreasing the amount of unstructured water. However, this broadening could be the result of new repulsive interactions between the three distinct anions forming the electrolyte. These new interactions may eliminate the hydrogen bonds between the first solvation shells of the MSA + HClO_4 system, thereby suppressing small peaks.

In contrast, Fig. 5B shows that, for 0.10 M MSA + NaCl in the range between 10^{-6} and 10^{-5} M, the OH_b peak remains nearly invariable with the addition of chloride, indicating that this anion does not adsorb in this range of concentrations. However, the OH_s peak broadens and shifts to lower potentials. A plausible explanation for this behaviour suggests that this is due to the enrichment of un-associated water molecules with dangling hydrogen bonds near the electrode surface owing to the presence of chloride [14]. The peak remains symmetric between the positive and negative scan directions, reinforcing the idea that chloride does not adsorb sufficiently in this concentration range to be detected in the voltammetric profile; otherwise, the process would be limited by the diffusion of chloride anions to the surface and would be clearly asymmetrical [14].

4. Discussion

The experimental results in this work show new voltammetric profiles for Pt(111) in acidic solutions when the following two conditions are fulfilled: i) the anions in the mixture do not specifically adsorb and ii) the anion concentrations are higher than those regularly studied (i.e. higher than 0.1 M concentration). A new “butterfly-shaped” voltammetric profile was found in the OH adsorption region, with several peaks. Based on previous reports [14,18], we propose that the distinct peaks correspond to OH adsorption from different water sources differing in the degree of hydrogen bonding. Hydrogen bonding is the most distinctive characteristic of water, which dynamically forms clusters of hydrogen-bonded molecules with different extensions in the liquid state. This behaviour strongly affects most of the physicochemical properties of water, conferring to it its very singular behaviour.

The elucidation of specific ion effects on solution properties has received significant attention in recent decades [27]. One of the earliest observations that led to categorising specific ion (cations and anions) effects was the influence of electrolytes on protein and carbohydrate solubility [27]. This led to the introduction of the Hofmeister series, which can be rationalised in terms of the effect of ions on the water structure. Accordingly, ions are classified as kosmotropic (or water structure-forming) and chaotropic (or water structure-breaking). Structure formation means that water is strongly bonded to anions (by

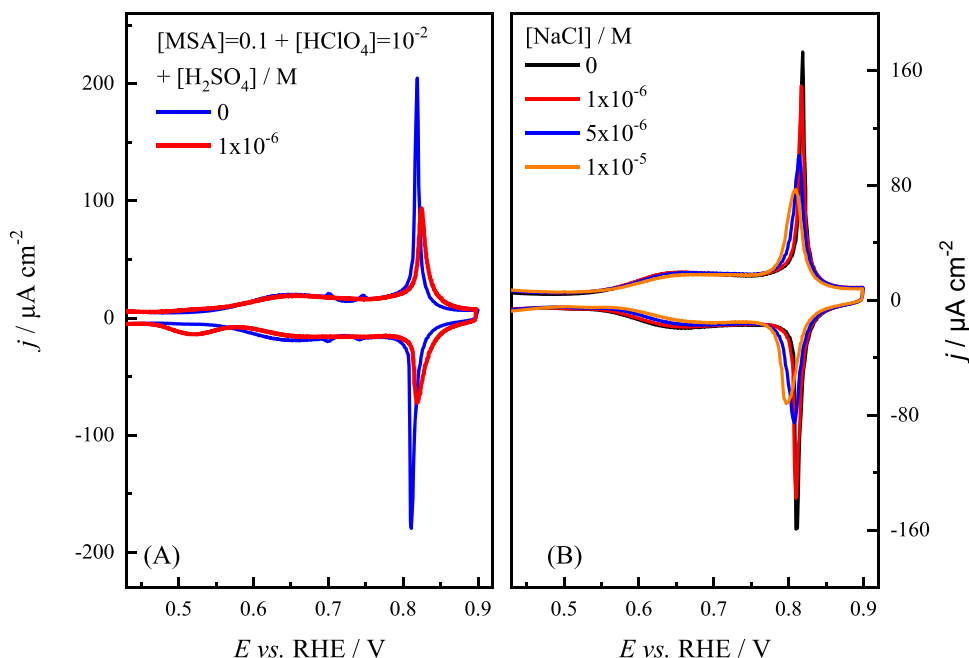


Fig. 5. Effect on cyclic voltammetric profiles of Pt(111) by addition of small amounts of (A) sulphate anion in 0.10 M MSA + 10^{-2} M HClO_4 pH 1.02 and (B) chloride anion in 0.10 M MSA, (pH 1.10). Scan rate=50 mV s^{-1} .

hydrogen bonds). Such structured water is distinguished from “free” or “bulk” unstructured water, which is weakly bonded to anions. It has been proposed that perchlorate anions have a weak interaction with water molecules, chloride has a negligible effect, and sulphate has a strong interaction with them [27]. However, the methanesulfonate anion has not yet been studied and classified; furthermore, shifts in the specific ion effects or even reversals in the Hofmeister series have been observed with changes in solvent, pH, temperature, counterion, or ion concentration. Currently, there is no reliable theory to quantitatively predict specific ion effects, although progress is being made [29].

The results obtained in this work indicate that an increase in the electrolyte anion concentration (whether methanesulfonate, perchlorate, sulphate, or chloride anions) affects the hydrogen bonding to i) anions-surrounding water molecules and ii) water-water networking. The unusual voltammetric profile obtained at a high MSA concentration, reported here for the first time, can be explained considering that the methanesulfonate anion has a remarkable ability as a hydrogen bond maker. The different mixtures of MSA + HClO₄ show diverse voltammetric peaks, the potentials of which depend on the anion concentration and pH. We hypothesise here that this voltammetric response is due to the existence of different clusters of water networks with various degrees of hydrogen bonding. Such a structure in the water network may be specific to the interfacial region and may also be affected by the electric field. The peak potential (or position of the voltammetric peak) is a measure of the adsorption energy and could be related to the strength of the hydrogen bonds, while the shape of the peak could indicate attractive (sharp) or repulsive (broad) interactions. Peak shifts to lower potentials indicate stronger hydrogen bond interactions that require lower adsorption energy to be absorbed, while peak shifts to higher potentials indicate a need for more energy to adsorb OH from a less structured hydrogen bond network. In this regard, the voltammetric electro-sorption response can be used as a “sensor” or “marker” of different hydrogen bond interactions. Under this hypothesis (but see below alternative explanations), as proposed in previous works, it is assumed that once the OH species are adsorbed on the surface, they are indistinguishable regardless of the degree of water organisation from which they came. This could explain why the total charge in the hydroxyl adsorption region remains virtually unchanged, although the shape of the isotherm reorganises according to the composition of the electrolyte.

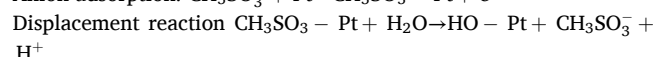
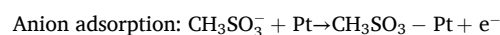
Under the hypothesis that the distinct peaks correspond to OH adsorption from different water sources differing in the degree of hydrogen bonding, we have two possibilities: i) if $E_{\text{OH}_b/(\text{H}_2\text{O})_b}^0 > E_{\text{OH}_s/(\text{H}_2\text{O})_s}^0$, then a double-layer region is expected; but if ii) $E_{\text{OH}_b/(\text{H}_2\text{O})_b}^0 < E_{\text{OH}_s/(\text{H}_2\text{O})_s}^0$, then an immediate transition from a OH_b-covered surface to a OH_s-covered surface is predicted, with no double-layer region (analogue to what is proposed on [30]). The voltammetric profiles at high MSA concentration seem to indicate a considerable double-layer (Fig. 4A and B) between 0.65 and 0.8 V vs RHE, which supports that the two different structured water have significantly different standard potentials. Following the ideas in [30], computational studies would be useful for validating this point.

In the explanations given above, we have considered that the effect of the anion on the structure of water is the main factor defining the voltammetric profile in the OH adsorption region. However, in the reaction $\text{H}_2\text{O} \rightleftharpoons \text{OH}_{\text{ads}} + \text{H}^+ + \text{e}^-$, there are two species that can be affected by the interaction with the anion. The effect on the water molecule is the one considered above, but anions could also interact with OH_{ads}, affecting its chemical potential. The interaction of the anion with OH_{ads} could be direct through hydrogen bonding with the oxygen atoms on the anion, or the anion could affect the nature of the water molecules bonded to the OH_{ads}. The latter would be similar to the second shell effect mentioned above for the structure of water. Previous studies have suggested that the perchlorate anion interacts with OH_{ads}[31]. A similar interaction can also be expected between methanesulfonate and adsorbed OH. At this point, separating both possibilities requires

computational studies including sufficient explicit water molecules in order to correctly describe these effects.

Another explanation for the existence of sharp peaks is the existence of a disorder-order phase transition. This phenomenon is well established for other adsorption processes that also generate sharp peaks or spikes, such as sulphate adsorption on Pt(111) or Au(111) [10]. Still, the peaks observed in the voltammograms of Pt(111) in MSA or HClO₄ are broader than those in sulfuric acid and involve too much charge to be considered a phase transition. Nevertheless, a quantitative study is required to definitely rule out the phase transition hypothesis.

Additionally, similar to sulfate, the specific adsorption of methanesulfonate anions on the platinum surface at high potentials cannot be discarded. However, there are significant differences in the behaviour of sulphate and methanesulfonate. First, the sharp peak OH_s is not observed in sulfuric acid solutions. If the possibility of methanesulfonate adsorption is accepted, this peak would imply a displacement of the anion by OH at high potentials. Such a displacement would be accompanied by a very small or non-charge flow, according to the following equations:



If MSA is adsorbed first and then replaced by OH_{ads}, the charge at the end would be the same as if only OH adsorption occurred, since the charge is a function of state, and in both cases, the surface is transitioning between an adsorption-free state to an OH-covered state. Hence, MSA adsorption could also explain the charge redistribution. As shown in Figures S2, S6, and S7, the charge for the first adsorption process below 0.7 V is about 40 μC cm⁻², much lower than the typical charge for other anion adsorption processes, which are between 80 and 100 μC cm⁻². However, if we make the comparison with sulphate anion adsorption, assuming that it forms a structure similar to that of sulphate and that the spike is the phase transition, a coverage around 0.2 would imply a charge of 48 μC cm⁻², similar to the one integrated with MSA. However, it would remain difficult to explain with this interpretation the following observations: i) the shift with the pH at constant MSA concentration, reported in Fig. 4A, ii) the synergic effect between MSA and HClO₄ evidenced in Fig. 3 and iii) the fact that no adsorption process is visible in 0.10 M MSA solutions, in striking contrast with the adsorption of other anions that are adsorbed in a broad range of concentration from 10⁻⁴ M upwards

Infrared spectroscopy (IR) is a valuable tool for studying hydrogen bonding in aqueous solutions. Water molecules have three vibrational modes that can be shifted by hydrogen bonding. The degree and nature of these shifts depend on the number and type of hydrogen bonds. Surface-enhanced infrared absorption spectroscopy (SEIRAS) allows for the study of interfacial water, revealing the influence of anions on the interfacial water structure. The effect of anions on interfacial water vibrations and hydrogen bonding has been investigated using this approach. Thus, a sharp band at 3610 cm⁻¹ was identified for water adsorbed on thin layers of gold in the presence of perchlorate and chloride anions. This band was interpreted as a signal of the presence of non-coordinated OH moieties adjacent to the anion [32], caused by the structure-breaking effect of chloride and perchlorate. The addition of sulphate, an anion classified as kosmotropic or structure-making, caused the disappearance of this band and a shift of the OH stretching to 3450 cm⁻¹ [32,33]. These precedents demonstrate that anions play a significant role in the structuring of interfacial water, thereby supporting the hypothesis presented here that varying degrees of water network formation lead to distinct voltammetric profiles. Previous spectroscopic studies [18,20] have also attempted to identify if methanesulfonate adsorbs on Pt surfaces. The absence of bands attributed to adsorbed methanesulfonate indicates that this anion does not specifically adsorb.

However, small amounts of adsorbed anions below the detection limit cannot be discarded.

In addition, DFT and molecular dynamics (MD) calculations could help describe the hydrogen bonds of anions-surrounding water molecules and water-water networking. Thus far, using DFT, it has been determined that the polarisability of MSA and the methanesulfonate anion is influenced by the number of hydrogen bonds in the water surrounding the clusters [23,24]. Clusters with up to three water molecules are dominated by strong hydrogen bond interactions with MSA. To date, existing DFT studies have been limited, including only up to five water molecules that solvate anions. The continuation of this work will require the use of DFT and MD to measure the strength of anion-water and water-water hydrogen bonds, and we are already working in this direction.

So far, it is known that, for anions, changes in the hydrogen bond network can be quantified by considering site-specific electrostatic interactions [29]. These interactions are considered key to understanding different properties such as viscosity or surface tension. Viscosity or surface tension is a good marker of changes in the geometry and strength of the hydrogen-bonded network because the interaction energy between a hydrated anion and a solvating water molecule and its enthalpy of hydration are correlated by site-specific electrostatic interactions [29]. The viscosity or surface tension of a solvent generally increases with the addition of salt or a decrease in temperature. Therefore, an increase in viscosity or surface tension indicates a higher proportion of hydrogen bonds [28,34]. As a perspective for future work, additional experiments could be performed with lower temperatures or even supercooled water or solutions with a highly concentrated salt content, generating an increase in the ratio of structured/unstructured water that should be correlated with changes in the voltammetry of Pt(111) in the OH adsorption region. In addition, thermodynamic analysis [35] of voltammetric profiles recorded at different temperatures could provide bond energies. These are affected by the double-layer properties, and they could reflect the effect of anions on the interfacial water network. As a very first step in this direction, Figure S8 in the Supporting Information shows that a temperature decrease to ca. 273 K in 0.10 M MSA solution triggers the emergence of the new peaks, while the voltammogram recorded at 294 K in the same electrolyte does not show these peaks. This finding could indicate a higher proportion of hydrogen bonds in the water network with decreasing temperature.

It is also worth pointing out recent studies that investigated the specific ion effect of alkaline cations in the hydrogen evolution process [36,37] and tetraalkylammonium cations [38] in the hydrogen evolution process. Structure-making Li^+ ions facilitate hydrogen evolution, while structure-breaking K^+ and Cs^+ ions hinder the reaction. The effect of the ions on the interfacial water was followed by SEIRAS and modelled using MD simulations. The effect of the cation on the water network structure is translated into the rate of hydrogen evolution through its effect on the reorganisation energy, which increases for the disordered water adlayers.

In summary, the new voltammetric responses of OHads can be considered evidence of both specific ion effects and rearrangements in the hydrogen bond dynamics induced in water molecules by methanesulfonate and perchlorate anions. Moreover, our results suggest that the coexistence of methanesulfate and perchlorate anions has a long-range structural effect on hydrogen bond rearrangement dynamics. In this regard, we tentatively correlated different voltammetric peaks with the contribution from water located in the first solvation shell or further away from the central ion. However, finding direct evidence for this presumption is not an easy task. The hydrogen bond network is strongly dependent on the environmental temperature, pressure, pH, ion concentration, and presence of dissolved oxygen. Additionally, it also depends on the interaction between ions, polarisation of water molecules in their vicinity, and ion-dependent changes in water clusters [34]. The interactions between water molecules far from the ions change significantly because of the cooperative effect between them. This complexity

is closely linked to the variation in properties induced by the presence of different ions. In order to gain a better understanding of the dynamics of hydrogen bond rearrangement, it is necessary to show the long-range distribution features of water molecules using macro (e.g. temperature effects) and microscopic (spectroscopic and DFT calculations) testing methods simultaneously. However, the high difficulty in conducting these tests, the lack of definition of the hydration shells, and the effective ranges affected by the anions make the study of the hydrogen bond rearrangement dynamics a difficult topic of research [34]. Additional experiments, including spectroscopic measurements and computational simulations, are in progress to completely elucidate the phenomena responsible for the new adsorption peaks.

5. Conclusions

In acidic media, a new response of OH adsorption with several peaks was found by increasing the methanesulfonate anion concentration (from 0.10 M to 0.30 M). It was found that perchlorate anions in the presence of methanesulfonate anions provide structured water, generating a synergic effect larger than that corresponding to the individual anions. It was confirmed that the sulphate anion has a structure-making effect while the chloride anion has a structure-breaking effect in MSA electrolytes, which agrees with previous studies using perchloric acid as the supporting electrolyte. The results of this work indicate that the OH adsorption response depends on the anion concentration and the specific hydrogen bonding anion-water (solvation shell) and water-water network. Our findings suggest that the distinct peaks correspond to OH adsorption from different water sources differing in the degree of hydrogen bonding, with the methanesulfonate anion being a remarkable hydrogen bond maker. However, possible interactions between anions and the surface and/or OHads, as well as possible interactions between water molecules and OHads, are not discarded. Further work is in progress to describe the interactions between hydrogen bonds and both acidic and basic electrolytes in various electrocatalytic materials.

CRedit authorship contribution statement

Victoria Gámez: Writing – original draft, Investigation. **Gabriel Melle:** Writing – review & editing, Investigation. **Víctor Climent:** Writing – original draft, Supervision, Conceptualization. **Rosa M. Arán-Ais:** Writing – review & editing, Supervision, Funding acquisition. **Enrique Herrero:** Writing – review & editing, Supervision, Funding acquisition. **Juan M. Feliu:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.electacta.2024.145494](https://doi.org/10.1016/j.electacta.2024.145494).

Data availability

Data will be made available on request.

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